WEEK 12-PROJECT REPORT

AUTOMOBILE ANALYSIS DASHBOARD USING POWER BI

1. Introduction

In Week 12, an Automobile Analysis Dashboard was created using Microsoft Power BI. The main aim of this project is to analyse automobile data and compare different car manufacturers based on price, engine size, horsepower and fuel efficiency.

Different visualizations such as KPI Cards, Scatter Chart, Column Charts and Tables were created to present the data in a clear and meaningful way.

2. Objective

- To analyse automobile data using Power BI.
- To compare the average price of different car manufacturers.
- To analyse engine size and horsepower.
- To understand highway mileage (MPG).
- To create an interactive and informative dashboard.

3. Tools Used

- Microsoft Power BI Desktop
- Automobile Dataset
- Power Query
- DAX Measures
- Data Visualization Tools

4. KPI Card Creation

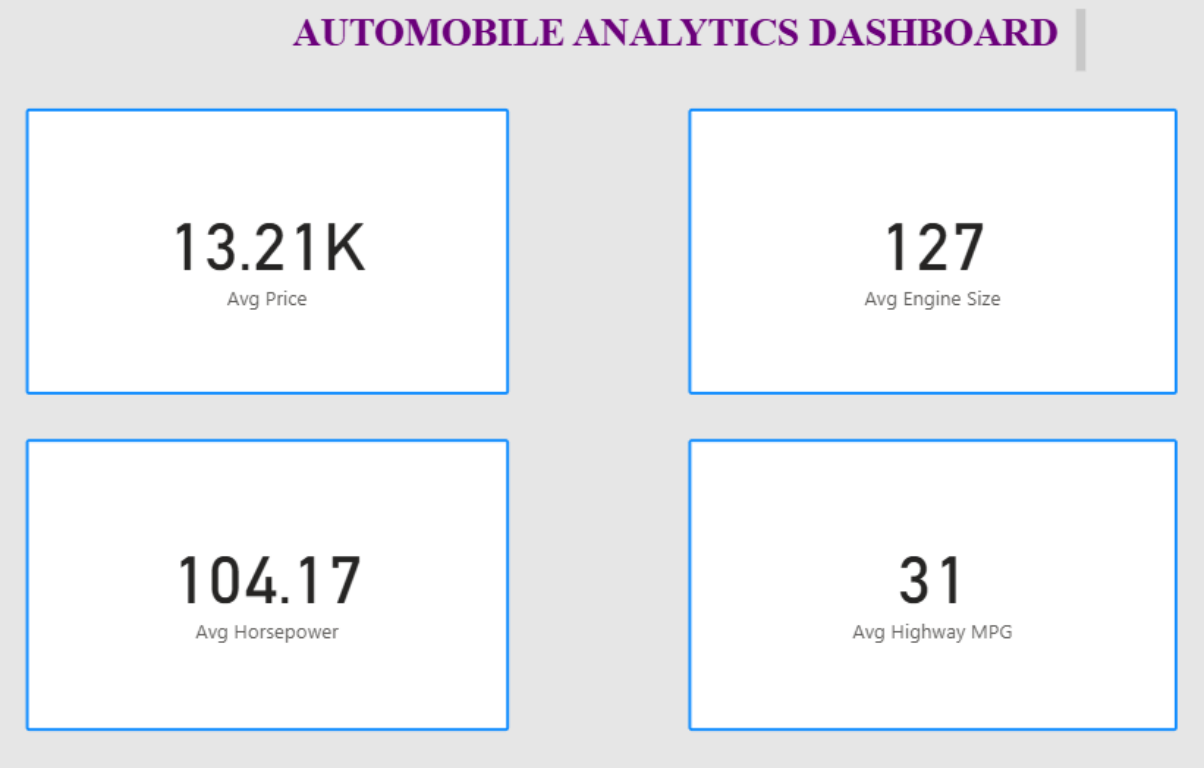
Four KPI Cards were created to display the important information about automobiles.

The following measures were used:

- Average Price
- Average Engine Size

- Average Horsepower
- Average Highway MPG

These cards provide a quick summary of the automobile dataset.

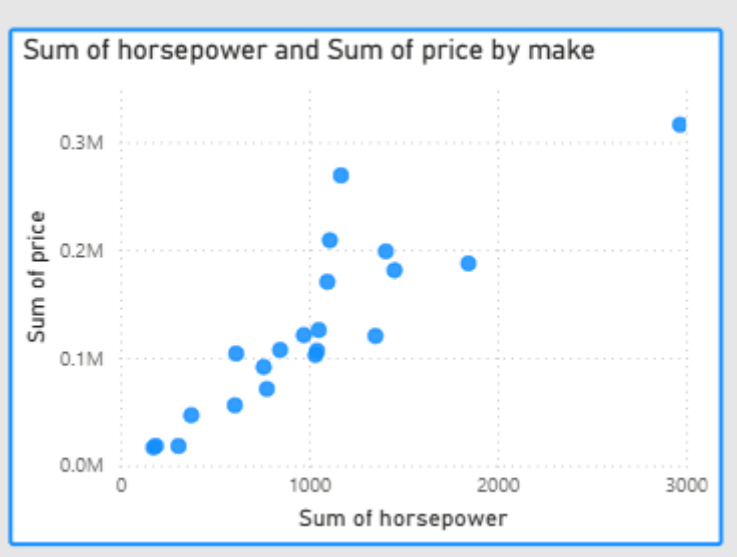


5. Scatter Chart Creation

A Scatter Chart was created to analyse the relationship between horsepower and price for different car manufacturers.

- X-axis: Sum of Horsepower
- Y-axis: Sum of Price
- Details: Make

This chart helps to understand the relationship between engine power and automobile prices.

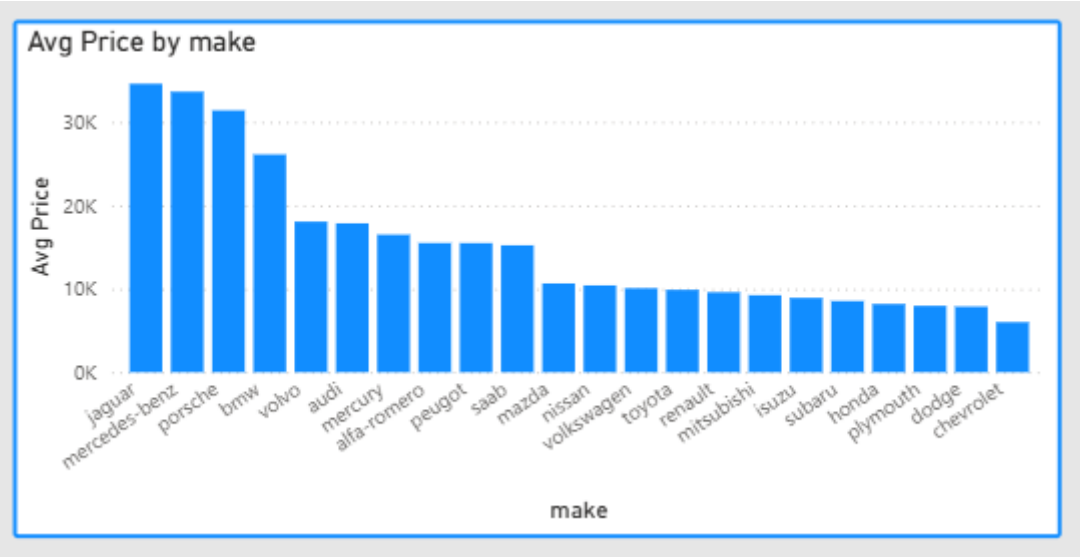


6. Column Chart Creation – Average Price by Make

A Clustered Column Chart was created to compare the average price of different car manufacturers.

- X-axis: Make
- Y-axis: Average Price

This chart helps to identify the differences in average prices among various car manufacturers.



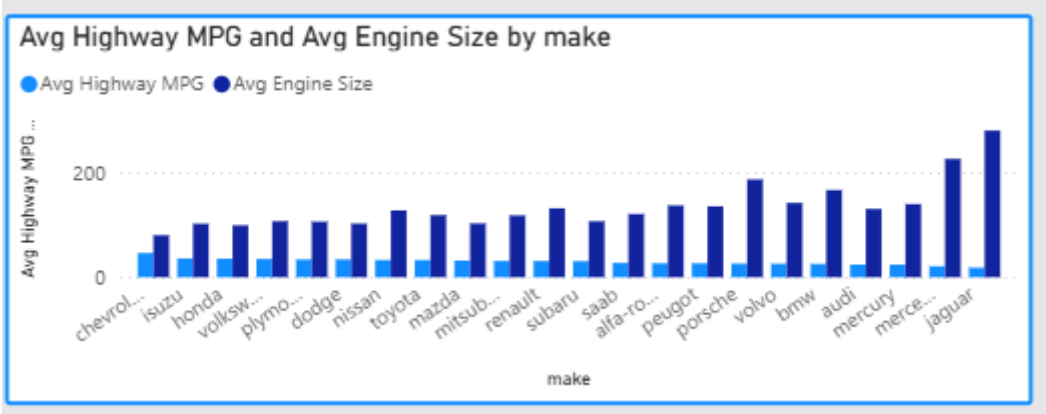
7.

Average Highway MPG and Average Engine Size by Make

A Clustered Column Chart was created to compare the average highway mileage and average engine size of different car manufacturers.

- X-axis: Make
- Y-axis: Avg Highway MPG and Avg Engine Size

This chart helps to analyse fuel efficiency and engine capacity across different car manufacturers.



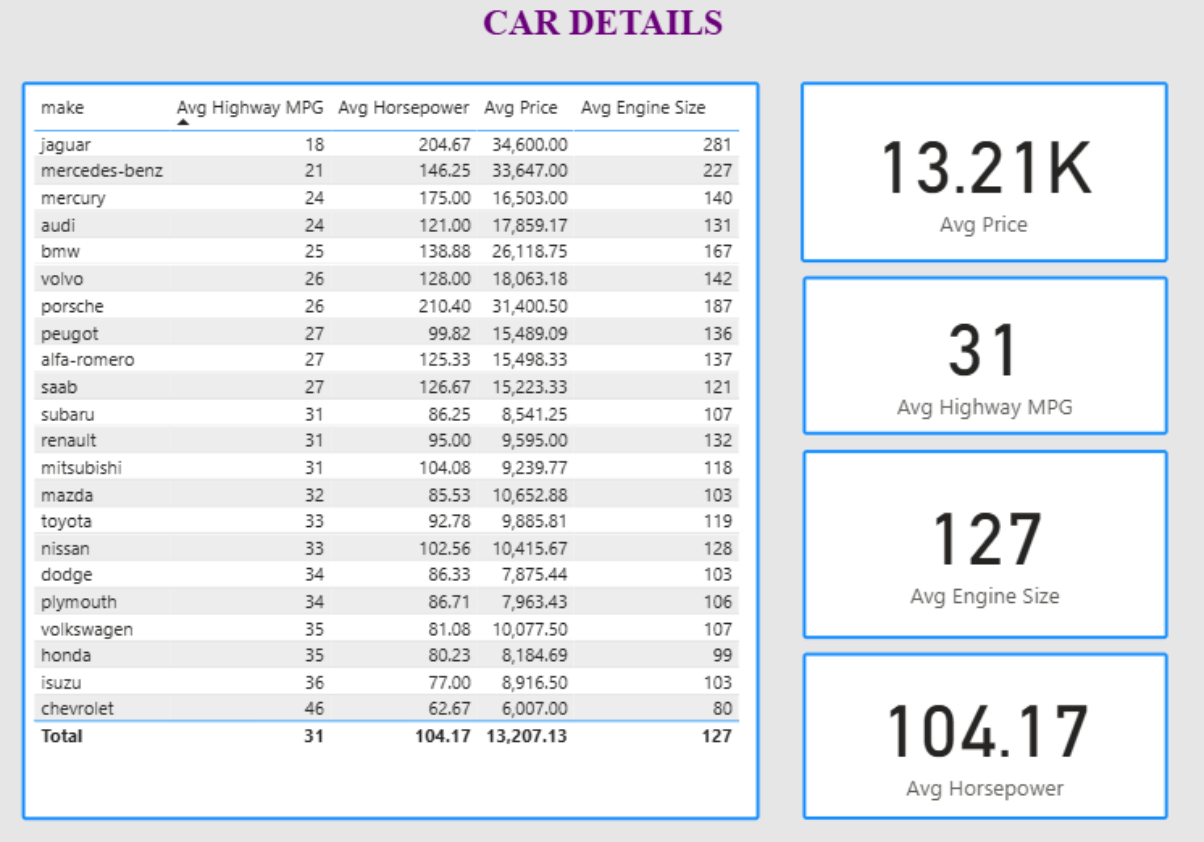
8. Table Creation – Car Details

A Table visual was created to display the automobile details based on different car manufacturers.

The following fields were included:

- Make
- Average Highway MPG
- Average Horsepower
- Average Price
- Average Engine Size

This table provides a detailed comparison of automobile specifications.



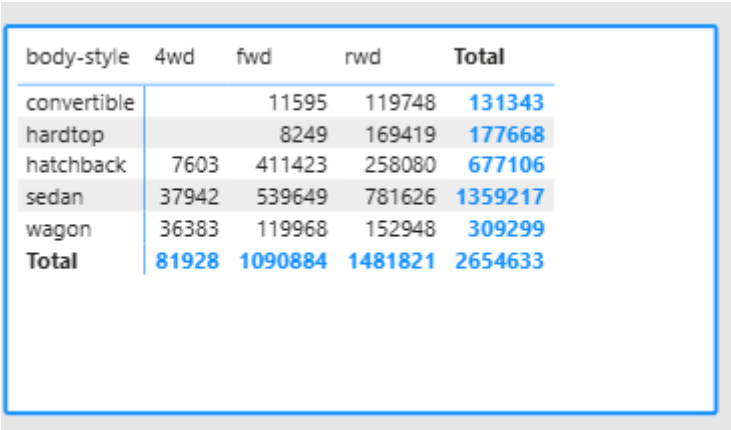
make	Avg Highway MPG	Avg Horsepower	Avg Price	Avg Engine Size
jaguar	18	204.67	34,600.00	281
mercedes-benz	21	146.25	33,647.00	227
mercury	24	175.00	16,503.00	140
audi	24	121.00	17,859.17	131
bmw	25	138.88	26,118.75	167
volvo	26	128.00	18,063.18	142
porsche	26	210.40	31,400.50	187
peugot	27	99.82	15,489.09	136
alfa-romero	27	125.33	15,498.33	137
saab	27	126.67	15,223.33	121
subaru	31	86.25	8,541.25	107
renault	31	95.00	9,595.00	132
mitsubishi	31	104.08	9,239.77	118
mazda	32	85.53	10,652.88	103
toyota	33	92.78	9,885.81	119
nissan	33	102.56	10,415.67	128
dodge	34	86.33	7,875.44	103
plymouth	34	86.71	7,963.43	106
volkswagen	35	81.08	10,077.50	107
honda	35	80.23	8,184.69	99
isuzu	36	77.00	8,916.50	103
chevrolet	46	62.67	6,007.00	80
Total	**31**	**104.17**	**13,207.13**	**127**

9. Matrix Visualization

A Matrix visual was created to analyse automobile prices based on body style and drive type.

- Rows: Body Style
- Columns: Drive Type
- Values: Sum of Price

This visualization helps to compare automobile prices across different body styles and drive types.



body-style	4wd	fwd	rwd	Total
convertible		11595	119748	131343
hardtop		8249	169419	177668
hatchback	7603	411423	258080	677106
sedan	37942	539649	781626	1359217
wagon	36383	119968	152948	309299
Total	**81928**	**1090884**	**1481821**	**2654633**

10. Dashboard Formatting and Design

The dashboard was formatted to improve its appearance and readability.

- Added suitable chart titles.
- Arranged the visualizations neatly.
- Used appropriate chart types.
- Displayed important information using KPI Cards.
- Organized automobile details using tables.

11. Result

The Automobile Analysis Dashboard was successfully created using Microsoft Power BI. The dashboard provides useful insights into car prices, horsepower, engine size and highway mileage. Different visualizations help to compare automobile manufacturers and understand the dataset effectively.

12. Conclusion

Through this project, I learned how to create KPI Cards, Scatter Charts, Clustered Column Charts, Tables and Matrix visualizations in Power BI. I also gained practical knowledge of data analysis, comparison and dashboard design.

Thus, the Week 12 Automobile Analysis Dashboard was successfully completed using Microsoft Power BI.